

Name: Mark Heien

Partner: Isaiah Ivy

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The Thermodynamics of Acetic Acid Dissociating Lab Report

Purpose.

The purpose of this lab report is to determine the equilibrium constant (K_a) for the dissociation of acetic acid in water and to calculate the corresponding thermodynamic parameters, including Gibbs Free Energy (ΔG°), which is dependent on the enthalpy change (ΔH°), the entropy change (ΔS°), and the temperature (T) of the reaction or process. The equation used to calculate ΔG° is given by: $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$. By obtaining K_a values at different temperatures, we can derive ΔH° , ΔS° , and ΔG° . Through experimental measurements and analysis using thermodynamic principles, we aim to deepen our understanding of the thermodynamics of this acid-base equilibrium reaction and demonstrate the application of these principles in predicting and interpreting the behavior of chemical systems.

Procedure

Two beakers were dried and marked with initials using a Sharpie. Approximately 150 mL of 1.0M acetic acid was placed in a 250 mL beaker with a watch glass. The beaker was then subjected to a hot-water bath at 40.0°C for 15-20 minutes. A cold-water bath was prepared in a 600 mL beaker by combining ice and cold tap water. Using a 250 mL beaker, 150 mL of 1.0M acetic acid was measured and chilled for 15-20 minutes with ice to maintain a temperature below 10°C. The temperature was monitored

until the probe was ready. The temperature probe cable was then plugged into a different LabQuest Mini channel port, and the room temperature was recorded. The acetic acid solutions were removed from the water baths, and the pH meter and temperature probe were placed into the solutions. The pH and temperature were recorded as they equilibrated to room temperature. Initial measurements were taken by submerging the probes into the solution. The pH and temperature were measured five times above room temperature with a 2-3°C interval and at least a 2°C difference. Measurements were taken once the acetic acid solution cooled below 35°C. The temperature of the cold-water bath was monitored using a probe, and ice was added as necessary to keep it below 10°C. Any excess liquid from the probes was shaken off into a beaker. The solutions were allowed to equilibrate for 10-15 minutes before taking pH and temperature measurements. The pH probe was rinsed and stored, and the temperature probe was rinsed with deionized (DI) water and dried with a Kimwipe. The above process was repeated for the cold-water solution, being careful to avoid water in the acetic acid. The beaker was dried, and pH and temperature measurements were taken every 2-3°C until five sets of data were recorded below room temperature, with a minimum 2°C difference between temperature readings. The pH and temperature probes were prepared with DI water and a Kimwipe, and then used to measure the pH and temperature of the acetic acid solution.

Data Tables.

Initial concentration of the acetic acid was 1.024M

Volume of 0.1M acetic acid was 150 mL

Temperature (°C)	pH
Hot water	
44.0	2.03
35.0	2.07
33.0	2.11
31.0	2.14
29.0	2.17
27.0	2.20
Cold water	
13.0	2.43
11.0	2.46
9.0	2.49
7.0	2.52
5.0	2.50

Calculations.

1. ICE table for the dissociation of acetic acid

$\text{CH}_3\text{COOH} \rightleftharpoons \text{H}^+ + \text{CH}_3\text{COO}^-$			
I	1.024M	0	0
C	-x	+x	+x
E	1.024M	x	x

The equilibrium expression for the acid dissociation constant, K_a , for acetic acid is: $K_a = \frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]}$

K_a for acetic acid

We need to first calculate the concentration of H⁺ ions in solution. We can do this using the pH measurement and the equation:

$$\text{pH} = -\log[\text{H}^+]$$

$$[\text{H}^+] = 10^{(-\text{pH})}$$

Concentration of acetic acid is 1.024

At 44.0°C, pH = 2.03

$$[\text{H}^+] = 10^{(-2.03)} = 9.33 \times 10^{(-3)} \text{ M.}$$

$$[\text{CH}_3\text{COO}^-] = x = 9.33 \times 10^{(-3)} \text{ M.}$$

At equilibrium, the concentration of CH₃COO⁻ is equal to [H⁺], which is 7.13×10⁽⁻³⁾ M.

$$K_a = [\text{H}^+] [\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}]$$

$$K_a = (9.33 \times 10^{(-3)})^2 / 1.024$$

$$8.51 \times 10^{(-5)}$$

Therefore, the value of K_a for acetic acid at 44.0°C is 4.87×10⁽⁻⁵⁾

At 35.0°C, pH = 2.07

$$[\text{H}^+] = 10^{(-2.07)} = 8.51 \times 10^{(-3)} \text{ M.}$$

At equilibrium, the concentration of CH₃COO⁻ is equal to [H⁺], which is 8.51×10⁻³ M.

$$K_a = [\text{H}^+] [\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}]$$

$$K_a = (8.51 \times 10^{-3})^2 / 1.024$$

$$K_a = 7.07 \times 10^{-5}$$

$$\text{At } 33.0^\circ\text{C } \text{pH} = 2.11$$

$$[\text{H}^+] = 10^{-2.11}$$

$$= 7.76 \times 10^{-3}$$

At equilibrium, the concentration of CH_3COO^- is equal to $[\text{H}^+]$, which is $7.76 \times 10^{-3} \text{ M}$.

$$K_a = [\text{H}^+] [\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}]$$

$$K_a = 7.76 \times 10^{(-3)^2} / 1.024$$

$$K_a = 5.88 \times 10^{-5}$$

$$\text{At } 31.0^\circ\text{C } \text{pH} = 2.14$$

$$[\text{H}^+] = 10^{-2.14} = 7.24 \times 10^{-3} \text{ M}$$

At equilibrium, the concentration of CH_3COO^- is equal to $[\text{H}^+]$, which is $7.24 \times 10^{-3} \text{ M}$.

$$K_a = [\text{H}^+] [\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}]$$

$$K_a = 7.24 \times 10^{(-3)^2} / 1.024$$

$$K_a = 5.13 \times 10^{-5}$$

$$\text{At } 29.0^\circ\text{C: } \text{pH} = 2.17$$

$$[\text{H}^+] = 10^{(-\text{pH})} = 6.76 \times 10^{-3} \text{ M}$$

$$[\text{CH}_3\text{COO}^-] = [\text{H}^+] = 6.76 \times 10^{-3} \text{ M}$$

$$K_a = [H^+] [CH_3COO^-] / [CH_3COOH]$$

$$K_a = (6.76 \times 10^{-3})^2 / 1.024$$

$$K_a = 4.46 \times 10^{-5}$$

$$\text{At } 27.0^\circ\text{C: pH} = 2.20$$

$$[H^+] = 10^{(-\text{pH})} = 6.31 \times 10^{-3} \text{ M}$$

$$[CH_3COO^-] = [H^+] = 6.31 \times 10^{-3} \text{ M}$$

$$K_a = (6.31 \times 10^{-3})^2 / 1.024$$

$$K_a = 3.89 \times 10^{-5}$$

$$\text{At } 13.0^\circ\text{C pH} = 2.43$$

$$[H^+] = 10^{(-\text{pH})} = 3.72 \times 10^{-3} \text{ M}$$

$$[CH_3COO^-] = [H^+] = 3.72 \times 10^{-3} \text{ M}$$

$$K_a = (3.72 \times 10^{-3})^2 / 1.024$$

$$K_a = 1.35 \times 10^{-5}$$

$$\text{At } 11.0^\circ\text{C pH} = 2.46$$

$$[H^+] = 10^{(-\text{pH})} = 3.47 \times 10^{-3} \text{ M}$$

$$[CH_3COO^-] = [H^+] = 3.47 \times 10^{-3} \text{ M}$$

$$K_a = (3.47 \times 10^{-3})^2 / 1.021$$

$$K_a = 1.17 \times 10^{-5} \text{ M}$$

$$\text{At } 9.0^\circ\text{C: pH} = 2.49$$

$$[\text{H}^+] = 10^{(-\text{pH})} = 3.24 \times 10^{(-3)} \text{ M}$$

$$[\text{CH}_3\text{COO}^-] = [\text{H}^+] = 3.24 \times 10^{-3} \text{ M}$$

$$K_a = [\text{H}^+] [\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}]$$

$$K_a = (3.24 \times 10^{-3})^2 / 1.021$$

$$K_a = 1.02 \times 10^{-5} \text{ M}$$

$$\text{At } 7.0^\circ\text{C: pH} = 2.52$$

$$[\text{H}^+] = 10^{(-\text{pH})} = 3.02 \times 10^{(-3)} \text{ M}$$

$$[\text{CH}_3\text{COO}^-] = [\text{H}^+] = 3.02 \times 10^{-3} \text{ M}$$

$$K_a = [\text{H}^+] [\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}] \quad K_a = [(3.02 \times 10^{-3})^2 / 1.021$$

$$K_a = 8.91 \times 10^{-6} \text{ M}$$

$$\text{At } 5.0^\circ\text{C: pH} = 2.50$$

$$[\text{H}^+] = 10^{(-\text{pH})} = 3.16 \times 10^{(-3)} \text{ M}$$

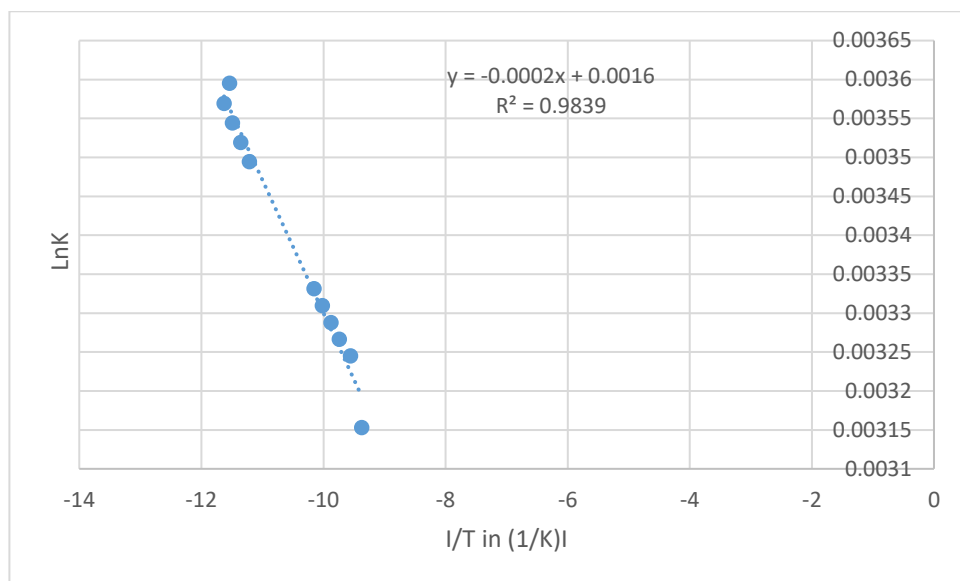
$$[\text{CH}_3\text{COO}^-] = [\text{H}^+] = 3.16 \times 10^{-3} \text{ M}$$

$$K_a = [\text{H}^+] [\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}]$$

$$K_a = (3.16 \times 10^{-3})^2 / 1.024$$

$$K_a = 9.77 \times 10^{-6} \text{ M}$$

A Van't Hoff plot of $\ln K$ versus $1/T$



The line equation with error is $y = -0.0002x + 0.0016$

Enthalpy change ΔH°

Slope (m) = $-0.0002 \text{ J/mol} \cdot \text{K}$.

Y-intercept (b) = 0.0016

Gas constant (R) = $8.314 \text{ J/mol} \cdot \text{K}$

$$\Delta H^\circ = -8.314 \text{ J/mol} \cdot \text{K} \times (-0.0002 \text{ KJ/mol}) \Delta H^\circ$$

Enthalpy change (ΔH°) is 0.0016628 KJ/mol.

Standard entropy change (ΔS°)

$$\Delta S^\circ = (-R) \times (\text{intercept of the Van't Hoff plot})$$

$$\Delta S^\circ = 0.0016 \times (-8.314 \text{ J/mol})$$

Entropy change (ΔS°) is $-0.0133024 \text{ KJ/mol} \cdot \text{K}$

Gibbs free energy change (ΔG°)

$$\Delta H^\circ = 0.0016628 \text{ KJ/mol}\cdot\text{K}$$

$$\Delta S^\circ = -0.0133024 \text{ KJ/mol}\cdot\text{K}$$

$$T = 24 + 273.15 = 297.15 \text{ K}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\Delta G^\circ = 0.0016628 \text{ KJ/mol} - (297.15 \text{ K} \times -0.0133024 \text{ KJ/mol}\cdot\text{K})$$

$$\Delta G^\circ = 0.0016628 \text{ kJ/mol} + 23.956 \text{ kJ/mol}$$

$$\Delta G^\circ = 23.9576 \text{ kJ/mol}$$

The standard free energy change at 24°C is 3.9576 kJ/mol.

Results

summary table with the temperature (in °C), pH measured, and the calculated K_a .

Temperature (°C)	pH	K_a
44.0	2.03	8.51×10^{-5}
35.0	2.07	7.07×10^{-5}
33.0	2.11	5.88×10^{-5}
31.0	2.14	5.13×10^{-5}
29.0	2.17	4.46×10^{-5}
27.0	2.20	3.89×10^{-5}
13.0	2.43	1.35×10^{-5}
11.0	2.46	$1.17 \times 10^{-5} \text{ M}$
9.0	2.49	$1.02 \times 10^{-5} \text{ M}$

7.0	2.52	$8.91 \times 10^{-6} \text{ M}$
5.0	2.50	$9.77 \times 10^{-6} \text{ M}$

A 7R×3C (with heading) with the temperature (in K), 1/T (in K⁻¹), and ln K_a

Temperature (K)	1/T	LnK _a
44.0	0.003153	-9.37168
35.0	0.003245	-9.55706
33.0	0.003266	-9.74137
31.0	0.00331	-9.87782
29.0	0.003332	-10.0178
27.0	0.003495	-10.1545
13.0	0.003519	-11.2128
11.0	0.003544	-11.3559
9.0	0.00357	-11.4931
7.0	0.003595	-11.6283

LINEST table

1/K	LnK
0.003153	-9.37168
0.003245	-9.55706
0.003266	-9.74137
0.003288	-9.87782
0.00331	-10.0178
0.003332	-10.1545
0.003495	-11.2128

0.003519	-11.3559
0.003544	-11.4931
0.00357	-11.6283
0.003595	-11.5362
$y = -0.0002x + 0.0016$	

A summary Table with values for ΔH° , ΔS° , and ΔG° and Standard deviation and the percent relative uncertainty for ΔH° and ΔS° , as well as Percent error for ΔG°

		Standard Deviation	Percent Relative Uncertainty	Percent Error
ΔH°	0.0016628 KJ/mol.	0.000747 KJ/mol.	44.89%.	
ΔS°	-0.0133024 KJ/mol·K	0.001935 KJ/mol·K	14.54%	
ΔG°	23.9576 kJ/mol.			9.91%

Discussion:

From the results, the standard deviation for ΔH° and ΔS° was moderately small, indicating that the data points for these parameters clustered tightly around the mean value. However, the percent relative uncertainty for ΔH° was quite high (44.89%), suggesting a significant level of uncertainty associated with this parameter. In contrast, the percent relative uncertainty for ΔS° was lower (14.54%), demonstrating a higher degree of confidence in the obtained value. The percent error for ΔG° was 9.91%, which fell within an acceptable range. This implies that the observed value of ΔG° (23.9576 kJ/mol) was 9.91% lower than the expected value (26.6 kJ/mol). These findings indicate the presence of some sources of error or uncertainty in the measurement or calculation of ΔG° , resulting in a lower-than-expected value.

These sources of error may have arisen from systematic errors related to equipment calibration and environmental factors, as well as human errors stemming from mistakes during the experimental procedure. To mitigate these errors and improve the experimental results, it is recommended to implement calibration and quality control measures, repeat the experiments, and control environmental factors. By implementing these suggestions, it would be possible to enhance the precision and accuracy of the obtained results, thereby improving the overall quality of the experiment.

Conclusion:

In conclusion, the dissociation of acetic acid under standard state conditions was found to be an endothermic process. The ΔH° value of 0.0016628 ± 0.000747 kJ/mol indicates that energy is required to break the bonds between the molecules of acetic acid. The ΔS° value of -0.0133024 ± 0.001935 kJ/mol·K suggests that the process is not favorable from an entropy perspective, as there is a decrease in the disorder of the system. Finally, the ΔG° value of 23.9576 kJ/mol with a percent error of 9.91% indicates that the process is non-spontaneous under standard conditions. Overall, these results suggest that the dissociation of acetic acid is not a spontaneous or favorable process under standard state conditions.